

CRT Chapter 9

1. Has an index value of 2

2. `int quantities[20];`

3. `double[ ] heights = {1.65, 2.15, 4.95}`

4.

```
for (int grade : grades) {  
    System.out.println(grade);  
}
```

6. When you pass a whole array to a method, the method gets access to the actual array, so it can view or change the elements inside it. But when you pass a single array element, only the value is passed, so changing it inside the method does not affect the original array element.

7. Offset array indexes are used when a range should be shifted to correlate to a lower range of index values. This keeps it aligned with real-world numbering while still using valid array indexes.

8. Output would be “ i “

10. A dynamic array would be a better structure choice for an online shopping cart. Allowing users to add or remove items at any time.

11. `ArrayList indexOf()` finds where an item is in the list. It looks through the list and uses `equals()` to compare items. If it finds a match, it returns the position (index). If it doesn't find the item, it returns -1.

12. Use `equals()` to compare wrapper class values because it checks the actual data inside the objects, not their object references.